5 Moist thermodynamics: Multi-component moist pseudo-adiabat, moist convection

5.1 Two-component moist (pseudo)-adiabat

In reality, the vast majority of planetary atmospheres are composed of multiple components, including both condensible and non-condensible species (for the temperatures and pressures of interest). Here we consider one limiting case of interest of an atmosphere with one condensible species alongside a background non-condensible species. This is relevant most notably to H_2O on Earth, alongside CH_4 on Titan and several condensibles on the gas and ice giants (including NH_3 and H_2O on Jupiter and Saturn and additionally CH_4 on Neptune, in which cases the fact that the background H_2 is lighter than the condensibles is a complicating factor not considered in detail here).

One can derive the generalised moist adiabat for a single condensible by generalising the first law of thermodynamics to two species, one that can condense with changes in temperature and pressure and one that cannot condense. To do so, we must make the following *assumptions:*

1. Rainout of condensate (i.e., condensate is instantaneously removed from the atmosphere upon formation), implying that we don't have to track the effect of condensate on the density or specific heat of an atmospheric parcel.

2. The atmosphere follows the ideal gas law, which is a reasonable approximation for the Earth-like pressures and temperatures of interest.

3. $\rho_{\rm sat} < \rho_{\rm condensate}$, consistent with our assumption of rainout above as well as our previous simplified form of Clausius-Clapeyron in Equation (4.3).

The first law of thermodynamics for a system with N_d moles of dry gas and $N_{\rm sat}$ moles of condensible gas at its saturation vapour pressure is

$$C_{p,d}dT + C_{p,c}dT - Vdp + LdN_{\rm sat} = \delta Q, \tag{5.1}$$

where $C_{p,d}$ is the heat capacity of the dry atmospheric component, $C_{p,c}$ is the heat capacity of the condensible vapour, V is the volume, and $dN_{\rm sat}$ is the number of moles of condensible vapour that condense when the system experiences a given temperature change dT and pressure change dp. Let us re-write Equation (5.1) using specific heat capacity (to be more consistent with the rest of our notes as well as the Pierrehumbert textbook), as in Pierrehumbert Eq. (2.28),

$$m_d c_{p,d}dT + m_c c_{p,c}dT - \frac{m_d}{\rho_d}dp_d - \frac{m_c}{\rho_c}dp_c + Ldm_c = (m_d + m_c)\delta q = 0, \tag{5.2}$$

where we have expanded the Vdp term for each individual species with $V = m/\rho$ and taken $\delta q = 0$[25] for an adiabat. Note that we have not included a term for heat storage in the

[25] I apologise for using q as both specific energy and mass mixing ratio – this term will be set to zero so it's irrelevant.

condensed phase in the air parcel, consistent with our assumption of rainout throughout this and the previous single-component derivation.

In the first problem sheet, you will derive this two-component moist adiabatic lapse rate – stated below in specific (per mass) units (equivalent to Eq. 2.33 of Pierrehumbert, 2010):

$$\frac{d\ln T}{d\ln p_d} = \frac{R_d + \frac{L}{T} q_{\rm sat}}{c_{p,d} + q_{\rm sat}\left(c_{p,c} - \frac{L}{T} + \frac{L^2}{R_c T^2}\right)}. \tag{5.3}$$

5.2 Dilute two-component moist adiabat

An important simplified form of the two-component moist adiabat is the *dilute* two-component moist adiabat – you will apply this in the first problem sheet, and we can derive it as follows.

The simplified dilute version of the first law of thermodynamics is

$$0 = c_{p,d} d\ln T - R_d d\ln p_d + \frac{L}{T} dq_c. \tag{5.4}$$

We can now make the assumption that the atmosphere is saturated when it begins to lie on the moist pseudo-adiabat – this implies that we can use Equation (4.3) to replace p_c by $p_{c,\rm sat}$ (i.e., the saturation vapour pressure of the condensible) as well as q_c by $q_{\rm sat}$. Thus, $d\ln p_{\rm sat} = \frac{L}{R_c T} d\ln T$, $q_{\rm sat} = \frac{p_{c,\rm sat}}{p_d}\frac{\mu_c}{\mu_d}$, with $p_{c,\rm sat}$ given by Clausius-Clapeyron. The change in saturation mass mixing ratio is then (Pierrehumbert, 2010 Eq. 2.31)

$$dq_{\rm sat} = \frac{\mu_c}{\mu_d} d\left(\frac{p_{c,\rm sat}}{p_d}\right) = \frac{\mu_c}{\mu_d}\frac{p_c}{p_d} d\ln\frac{p_c}{p_d} = q_{\rm sat}\left(d\ln p_{c,\rm sat} - d\ln p_d\right) = q_{\rm sat}\left(\frac{L}{R_c T} d\ln T - d\ln p_d\right). \tag{5.5}$$

We can then express

$$0 = c_p d\ln T - R_d d\ln p_d + \frac{q_c L^2}{R_c T^2} d\ln T - \frac{L q_c}{T} d\ln p_d. \tag{5.6}$$

Rearranging, we find the dilute moist adiabat

$$\frac{d\ln T}{d\ln p} = \frac{R_d + \frac{q_c L}{T}}{c_p + \frac{q_c L^2}{R_c T^2}}. \tag{5.7}$$

We can convert from q to ξ using $q = \xi\frac{\mu_c}{\mu_d}$. This version given below in the case of one dilute condensible species in per mole rather than per mass units:

$$\frac{d\ln T}{d\ln p} = \frac{R_u + \frac{L\xi}{T}}{c_p + \frac{L^2\xi}{R_u T^2}}. \tag{5.8}$$

In Equation (5.8), $\xi = p_{\rm sat}/p_d$ is the number (or volume) mixing ratio of the condensible gas, i.e., the ratio of the saturation vapour pressure to the partial pressure of the dry gas component p_d. Note that L and c_p in Equation (5.8) are also written per mole, rather than per mass – if they were per mass, the expression would include the molecular mass ratio of the condensible and dry species.

5.2.1 Discussion of dilute two-component moist adiabat

As stated above, Equation (5.8) is now the *dilute* limit of the moist (pseudo)-adiabat, where we have taken $d\ln p_d \to d\ln p$ (i.e., we assumed that there is a small enough abundance of condensible such that its partial pressure does not have a significant impact on the total atmospheric pressure), and assumed that $q_{\rm sat}$ is small[26]. This dilute limit is reasonable for Earth, but not for atmospheres which are composed of a condensible with high mass mixing ratio (e.g., steam atmospheres on exoplanets with a significant background non-condensible species).

Let's check the limits of Equation (5.8). First, note that as $\xi \to 0$ (which also corresponds to $q \to 0$), $\frac{d\ln T}{d\ln p} \to R/c_p$, which is the dry adiabat. Conversely, as $\xi \to \infty$ ($q \to \infty$), $\frac{d\ln T}{d\ln p} \to RT/L$, which is the expression for the one-species moist (pseudo)-adiabat that we derived in Equation (4.5).

Figure 5.1 (Figure 2.7 of Pierrehumbert, 2010) shows the moist adiabat for water vapour in Earth air for a 1 bar background atmosphere comparable to that of present-day Earth. As

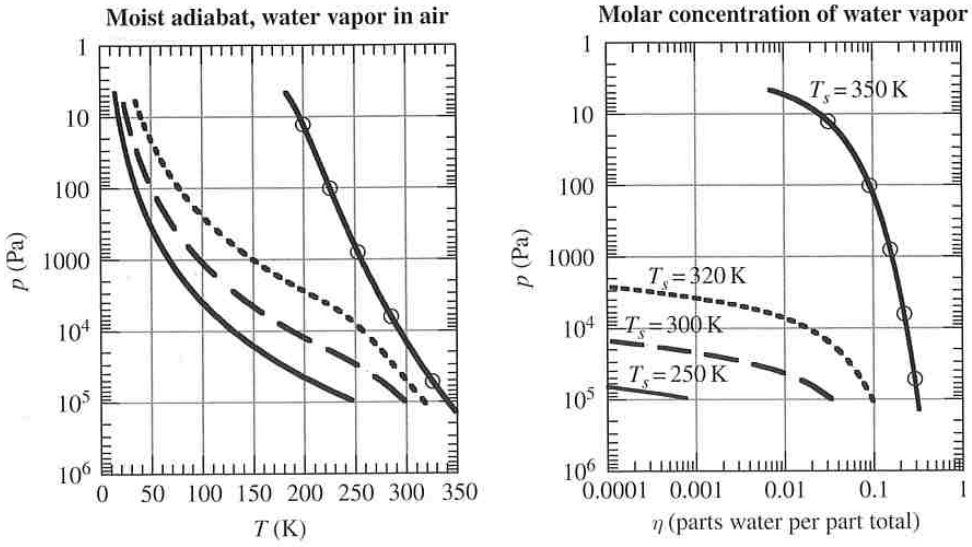


Figure 5.1: Moist adiabat for dilute saturated water vapour mixed with Earth air, where the dry component has a partial pressure of 1 bar at the surface. The left-hand panel shows the temperature profile, and the right shows the molar concentration of water vapour (ξ in these notes). Results are shown for varying surface temperature (linestyles). Figure adapted from Pierrehumbert (2010).

the surface temperature increases, the moist adiabat becomes shallower due to the additional latent heat release as the saturated parcels with higher molar concentration rise and condense. Note that the dilute limit begins to break down for sufficiently high surface temperatures, as water vapour is well-mixed with high molar ratios $\gtrsim 0.1$ throughout the troposphere.

Let's also compare the two-component moist adiabatic lapse rate to the dry adiabatic lapse rate $\frac{d\ln T}{d\ln p} = R/c_p$. We find that the moist adiabatic lapse rate is shallower than the dry

[26]This is equivalent to neglecting the two terms regarding the contribution of the moist air to the non-condensing entropy in the non-dilute version of the first law of thermodynamics.

adiabatic lapse rate if

$$\frac{R_u + \frac{L\xi}{T}}{c_p + \frac{L^2\xi}{R_u T^2}} < R_u/c_p,$$

equivalently,

$$\frac{1 + \frac{L\xi}{R_u T}}{1 + \frac{L^2\xi}{c_p R_u T^2}} < 1. \tag{5.9}$$

In the limit of large L, we find that this criterion implies $L/(Tc_p) > 1$, exactly the same condition as we had for the single-component moist adiabat.

Finally, note that the moist adiabat starts to deviate from that of a single-component gas when the dry terms in the numerator and denominator (R_u and c_p, i.e., not involving ξ) become important. Using the numerator alone, this deviation from a single-component moist adiabat occurs when $\xi \lesssim \frac{RT}{L}$ – for Earth, this occurs when $\xi \sim 0.05$. This is roughly an order of magnitude higher than the typical mixing ratio of water vapour on Earth, implying that the moist adiabat on Earth cannot be well represented by the single-component limit. Note that this also implies that the moist adiabat depends on ξ itself, which in turn depends strongly on pressure because ξ is exponentially sensitive to temperature through Clausius-Clapeyron. Thus, the moist-adiabatic gradient will converge toward a dry adiabatic gradient in the upper troposphere as $\xi \to 0$ due to the exponential decrease of saturation vapour pressure with decreasing temperature.

5.3 Moist stability criterion: conditional instability

We of course expect the atmosphere to be stable to convection if the environmental lapse rate dT/dz has a greater magnitude (is more negative) than the adiabatic lapse rate $(dT/dz)_{\rm ad}$. However, we have derived two "adiabatic" lapse rates – the dry adiabat, as well as the moist pseudo-adiabat. In reality, the stability of a parcel of gas depends on the environmental lapse rate relative to both the dry and moist adiabatic lapse rates, as well as whether the air in that parcel of gas is saturated.

Figure 5.2 shows the three possibilities for atmospheric stability given an environmental lapse rate $\Gamma = dT/dz$. These can be summarised as follows:

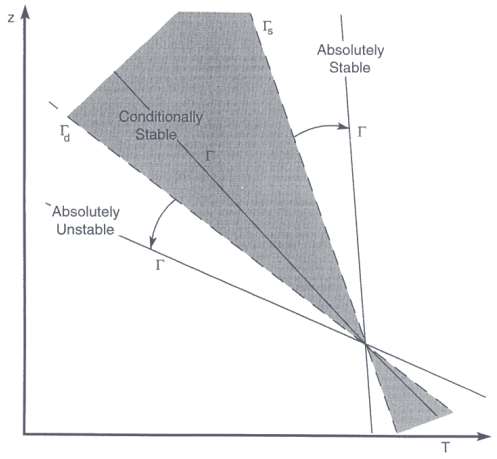


Figure 5.2: Vertical stability of moist air as a function of temperature and height for given values of the environmental lapse rate Γ. Figure adapted from Salby (1996).

(a) $\Gamma < \Gamma_{\rm dry}$: If the lapse rate is steeper (more negative) than the dry adiabat, the atmosphere is unstable to convection (i.e., adiabatic perturbations) regardless of its relative humidity. If the air is unsaturated, the air will follow the dry adiabat. If it is saturated, the air will follow the moist adiabat.

(b) $\Gamma > \Gamma_{\rm moist}$: If the lapse rate is larger (less negative) than the moist adiabat, then the atmosphere is absolutely stable to convection regardless of the relative humidity.

(c) $\Gamma_{\rm dry} < \Gamma < \Gamma_{\rm moist}$: In this case, the atmosphere is stable against dry convection but unstable to moist convection. This implies that unsaturated air parcels will return to their original position if perturbed along the dry adiabat. However, saturated air parcels (following the moist adiabat) will become buoyant and convect. This scenario is termed conditional instability, with the atmosphere considered to be *conditionally stable*.

The lapse rate in Earth's troposphere lies between the moist and dry adiabatic gradients, implying that our troposphere is conditionally stable. Meanwhile, stratospheres (by definition) are stable to convection as in case (b) above.

The fact that Earth's atmosphere as a whole is conditionally unstable is governed through multiple processes. In the tropics, a key process is moist convection itself (leading to localised regions of rapid ascent), as well as mass conservation requiring descending (and resulting relatively dry) regions that can cool via radiation to space (see Figure 5.3). Because the

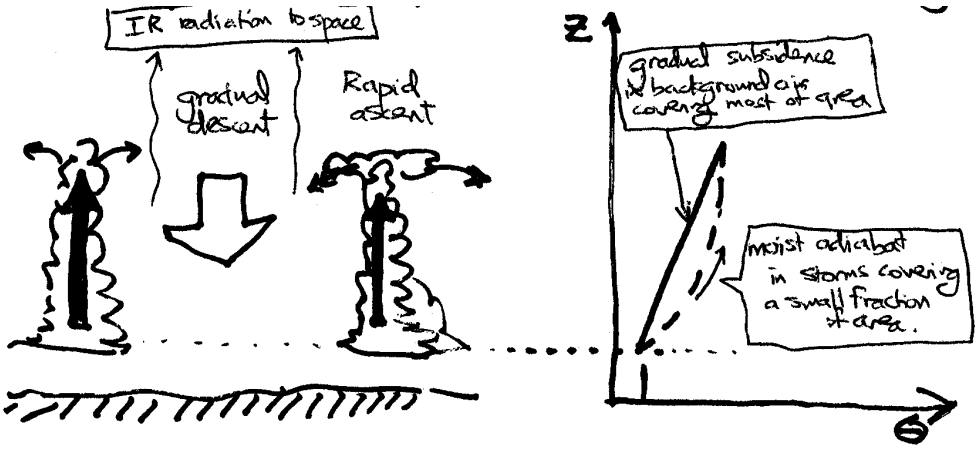


Figure 5.3: Schematic showing how moist convection interacts with the larger-scale (drier) environment, producing a conditionally unstable profile. Left: schematic showing smaller regions of strong moist convection (clouds) and larger drier descending regions radiatively cooling to space. Right: schematic potential temperature-height profile showing the typical profile in a moist region (dashed) and that in a dry region (solid). Figure courtesy Adam Showman.

background air radiates to space, its potential temperature θ decreases during subsidence, leading to a conditionally unstable profile. The difference between this profile and the moist adiabat implies that ascending saturated moist convective plumes have large buoyancy, leading to rapid ascent on the order of hours (compared to the subsidence over the course of

days to weeks in dry regions). This causes the convection on Earth to be highly asymmetric, with fast updrafts covering a small fraction of the total area, even in the relatively moist tropics.

5.4 Describing moist convection: LCL, LFC, LNB, moist CAPE

Figure 5.4 shows a typical temperature-altitude profile for a moist parcel (dashed line) compared to the dry adiabat (dotted line) and a typical background (environmental) profile (solid line). The point at which the parcel first condenses and moves from the dry adiabat

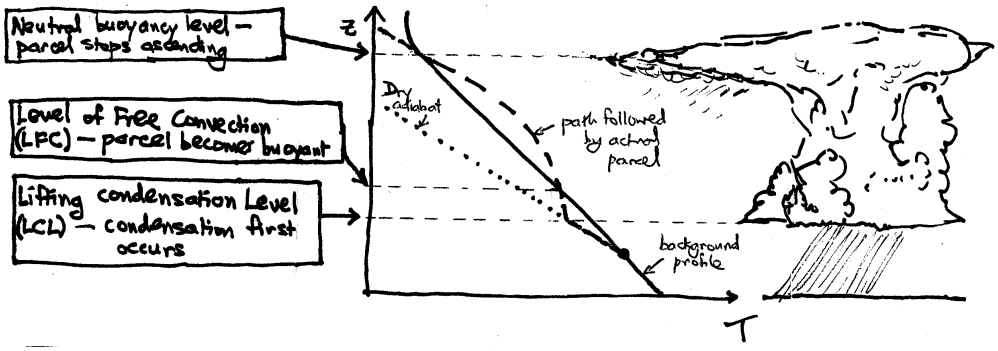


Figure 5.4: Annotated temperature-height profile showing the thermodynamic path followed by a moist parcel of gas as it ascends and condenses, forming a cloud (shown on the right). Figure courtesy Adam Showman.

(at lower altitudes) to the moist pseudo-adiabat (at higher altitudes) is termed the lifting condensation level, or LCL. This is where a cloud deck can begin to form (see sketch on the right), with corresponding rainout of moisture below the cloud deck. The saturated parcels then become buoyant (with respect to the background environmental profile) at the level of free convection, or LFC. At this point, the saturated parcels undergo rapid ascent up to the region where they dry out and the parcel's temperature profile crosses the environmental profile once again – this is the level of neutral buoyancy (LNB), where the parcel stops ascending.

Finally, note that moist CAPE can be calculated in much the same way as dry CAPE (Equation 3.14). However, the key difference is that the parcel's temperature for moist CAPE is calculated with respect to the moist rather than the dry adiabat. Similarly, the definition of moist LFC and LNB are equivalent to the dry version, except again the parcel is following the moist rather than the dry adiabat. The key difference in the moist case is that if the parcel begins unsaturated at low altitudes, it will then rise following the dry adiabat up to the LCL (as in Figure 5.4), above which it is lifted along the saturated moist adiabat until it reaches the LNB.

5.5 Some moisture-related definitions to know

5.5.1 Humidity metrics

There are two key humidity metrics that you will see used regularly – know the differences between them!

1. Specific humidity: This is the *specific concentration* of water vapour. The specific concentration is defined as the ratio of the mass of a given substance to the total mass of a parcel,
$$q_A \equiv \frac{\rho_c}{\rho_{\text{tot}}} = \frac{\mu_c}{\bar{\mu}} \frac{p_c}{p_{\text{tot}}} = \frac{\mu_c}{\bar{\mu}} \xi_c, \tag{5.10}$$
where $\bar{\mu}$ is the mean molecular weight of the mixture, ρ_{tot} is the total mass density of the parcel, and the subscript c represents the condensible species of interest (water vapour in this case).

2. Relative humidity: This is the degree of subsaturation, equivalent to the *saturation ratio*
$$\text{RH} = \frac{p_c}{p_{c,\text{sat}}}, \tag{5.11}$$
where p_c is the partial pressure of the condensible (water vapour in this case) and $p_{c,\text{sat}}$ is the saturation vapour pressure of the condensible (given by Clausius-Clapeyron). Note that the relative humidity is equal to the ratio of the volume or number mixing ratio of the condensible in a given mixture to the mixing ratio the substance would have if it were saturated. This is then of course different to the ratio of specific humidity to saturation specific humidity, because specific humidity is defined using mass rather than number or volume mixing ratio.

Another common quantity regarding moisture (sub)saturation that you may see (especially in more public/popular contexts) is the "dew point" or "frost point" depression, defined as $T - T_{\text{dew,frost}}(p_c)$. In this case, $T_{\text{dew,frost}}$ is the condensation temperature following Clausius-Clapeyron (Equation 4.6).

5.5.2 Virtual temperature

Virtual temperature is a quantity used to represent the temperature dry air needs to have the same pressure and density as moist air[27]. We know from ideal gas that dry air follows $p_d = \rho_d R_d T$, while that of the condensible follows $p_c = \rho_c R_c T$. The virtual temperature leverages this to write ideal gas relationship for moist air (the combination of dry and water vapour in this case) as
$$p = \rho R_d T_v, \tag{5.12}$$
where T_v is the virtual temperature. Using the ideal gas relationships for each component of the moist air, we can express virtual temperature as (Equation 2.64 of Holton & Hakim, 2013)
$$T_v = \frac{T}{1 - \frac{p_c}{p}\left(1 - \frac{R_d}{R_c}\right)}. \tag{5.13}$$
Note that $T_v \geqslant T$, and the difference is typically quite small – on the order of a few K. Regardless, these small differences can be critical to the conditional instability of a moist parcel.

[27] I *never* use this quantity because I find it confusing to mix multiple equations of state for an imaginary yet sometimes useful quantity, but you will see this quite often in papers and texts and should be familiar with how it is defined.